SIMATS ENGINEERING

Case Study 1: Medical Imaging & Surgical Robotics

COMPUTER VISION-ITA0513

Case Study: Keypoint Stability for Real-Time Surgical Tool Tracking

CO3-AT2-CASE STUDY

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Technical Executive Summary

The surgical robot must track a scalpel tip from a continuous endoscopic camera feed. The environment creates difficult visual conditions such as fluid occlusion, tissue deformation, reflections, glare, camera rotation, and scale changes. A reliable keypoint detector is therefore required.

Proposed solution:

SIFT (Scale-Invariant Feature Transform) is preferred over the Harris Corner Detector for tracking because SIFT provides greater robustness to image scaling and rotation. Harris is effective for detecting corners but its response is not inherently scale- or rotation-invariant. SIFT detects extrema across a scale-space and generates orientation-normalized descriptors, making feature matching more stable when the camera viewpoint or tool appearance changes.

Core Analysis & Methodology

1.1: Harris vs. SIFT

Harris Corner Detector

The Harris detector calculates image gradients and constructs the second-moment matrix:

$$M = \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$

The Harris response is:

$$R=\det(M)-k(\mathrm{trace}(M))^2$$

A large positive R indicates a likely corner.

Harris is computationally efficient and good at detecting strong corners, but its keypoint locations can change when the image is significantly scaled or rotated.

SIFT

SIFT searches for keypoints across multiple image scales using a Difference-of-Gaussians scale space:

$$D(x,y,\sigma)=L(x,y,k\sigma)-L(x,y,\sigma)$$

where:

$$L(x,y,\sigma)=G(x,y,\sigma)*I(x,y)$$

SIFT also assigns an orientation to each keypoint and constructs a descriptor relative to that orientation.

Which is more reliable?

SIFT is more reliable for this application.

The mathematical reason is that SIFT explicitly incorporates **scale σ** and **keypoint orientation** into its detection and description process. Therefore, a feature can be represented consistently even when the image undergoes rotation or scale changes.

Harris primarily uses local spatial gradients and does not inherently search across scale space.

Comparison

Feature	Harris	SIFT
Corner detection	Excellent	Excellent
Rotation robustness	Limited	High
Scale robustness	Limited	High

Descriptor	No standard descriptor	128-dimensional descriptor
Computational cost	Low	Higher
Real-time suitability	Very Good	Moderate
Tool tracking under viewpoint changes	Moderate	Better

Conclusion:

For a laparoscopic tool-tracking system where camera rotation and scale variation are expected, **SIFT is the stronger choice**, assuming the computational hardware can support its higher processing cost.

1.2: Filtering False Corners Caused by Glare

The standard Harris response is:

$R=\det(M)-k(\text{trace}(M))2$

A simple approach is to increase the detection threshold:

$R>T$

where T is the corner-response threshold.

However, simply using a fixed threshold may not be sufficient because surgical illumination can change significantly.

Adaptive Threshold

I would use an adaptive threshold based on the distribution of Harris responses:

$T=\mu R+\alpha\sigma R$

where:

- μR = mean Harris response
- σR = standard deviation
- α = experimentally selected factor

Then retain only:

$R(x,y)>T$

This rejects many weak or abnormal responses generated by changing illumination.

Additional Glare Filtering

A stronger engineering solution is to combine the Harris response with a brightness mask.

For example:

$R'(x,y)=R(x,y)\cdot W(x,y)$

where $W(x,y)$ reduces the response in highly saturated regions.

For an 8-bit image, pixels approaching the maximum intensity can be classified as possible glare:

$I(x,y)>T_g$

These regions can be excluded from keypoint selection.

Practical Pipeline

Endoscopic Frame

↓

Grayscale Conversion

↓

Harris Gradient Calculation

↓

Second-Moment Matrix M

↓

Harris Response R

↓

Adaptive Threshold

↓

Glare/Saturation Mask

↓

Valid Keypoints

This is preferable to merely increasing the Harris threshold because it specifically addresses **dynamic illumination artifacts**.

1.3: Validation Strategy

The descriptor should be tested under different illumination conditions while keeping the surgical tool and camera geometry as similar as possible.

Test Conditions

Create three controlled illumination groups:

1. Low-intensity illumination
2. Normal surgical illumination
3. Overexposed/high-intensity illumination

For each condition, capture multiple frames containing the scalpel tip.

Evaluation Procedure

For each frame:

1. Detect keypoints.
2. Extract SIFT descriptors.
3. Match descriptors between the reference frame and test frame.
4. Remove incorrect matches using a distance-ratio test and geometric verification.
5. Calculate stability metrics.

Metrics

1. Keypoint repeatability

Repeatability=Number of reference keypointsNumber of correctly repeated keypoints

A higher value indicates better stability.

2. Matching accuracy

Precision=TP+FPTP

where TP represents correct tool-tip matches and FP represents false matches.

3. Localization error

Measure the distance between the detected tool-tip position and the manually annotated ground-truth position:

E=(xd−xg)2+(yd−yg)2

where (xd,yd) is the detected position and (xg,yg) is the ground truth.

Acceptance Criteria

The SIFT-based system should demonstrate:

- High keypoint repeatability between lighting conditions.
- High percentage of correct descriptor matches.
- Low tool-tip localization error.
- Low false-match rate under glare.
- Stable performance across multiple illumination levels.

Validation Flowchart

Reference Surgical Frame
↓
Extract SIFT Keypoints
↓
Generate SIFT Descriptors
↓
Illumination
Changes
↓

Low / Normal / High
Illumination
↓
Extract Descriptors
↓
Match Keypoints
↓
Geometric Verification
↓
Calculate Performance
↓
Repeatability / Localization Error

Engineering Trade-offs & Limitations

Harris

Advantages

- Computationally inexpensive.
- Simple implementation.
- Fast for real-time processing.
- Suitable when scale and viewpoint changes are small.

Limitations

- Less robust to scale changes.
- Sensitive to illumination changes.
- Produces false responses around strong reflections.

SIFT

Advantages

- Scale-invariant.
- Rotation-invariant.
- Strong descriptor for feature matching.
- More reliable under camera movement and appearance changes.

Limitations

- More computationally expensive than Harris.
- Higher memory and processing requirements.
- Glare and complete occlusion can still cause feature loss.

Engineering Solution

For a surgical robot, I would use **SIFT for robust tracking**, combined with:

- Adaptive illumination normalization.
- Saturation/glare masking.
- Temporal tracking between consecutive frames.
- Geometric verification of matches.
- A fallback mechanism when the tool tip is temporarily occluded.

This provides better reliability than depending only on a raw corner detector.

Verification & References

The system can be validated using manually annotated scalpel-tip positions across multiple video sequences. Performance should be reported using **repeatability, matching precision, localization error, false-positive rate, and processing time per frame**.

For implementation, OpenCV provides documentation for both Harris corner detection and SIFT-based feature extraction.

Conclusion

For the laparoscopic surgical-tool tracking problem, **SIFT is preferable to Harris** because its scale-space detection and orientation-normalized descriptor provide stronger robustness to **camera rotation and scale changes**. Harris can still be useful when computational speed is the primary requirement, but SIFT provides better feature stability for the challenging surgical environment described in the case study.